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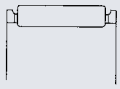
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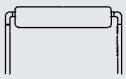
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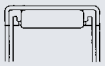


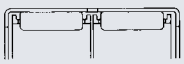
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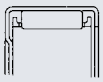
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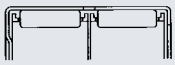

K
K· ·S
KJ· ·S

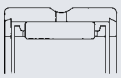

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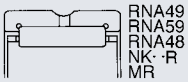

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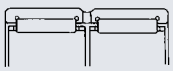

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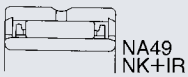

BK


BK· ·ZWD


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NK

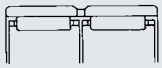

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NK· ·R
MR

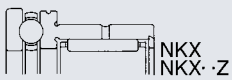

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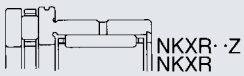

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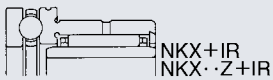

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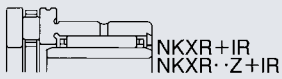

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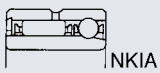

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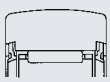

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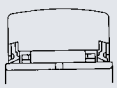

NKXR· ·Z
NKXR


NKX+IR
NKX· ·Z+IR


NKXR+IR
NKXR· ·Z+IR


NKIA

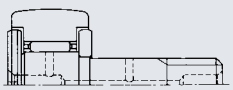

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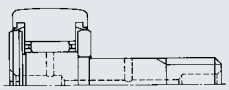

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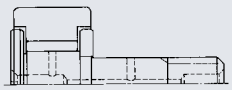

NATR


NATR· ·LL


NATV
NACV


KR


KR· ·LL


KRV
CRV


KRV· ·LL
CRV· ·LL


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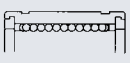

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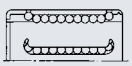

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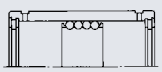

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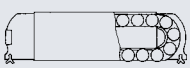

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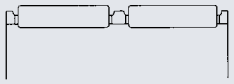
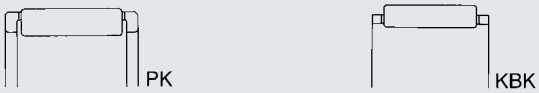
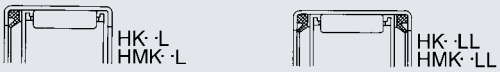
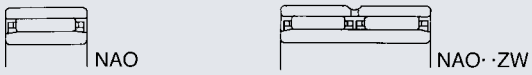
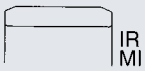
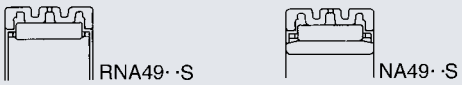
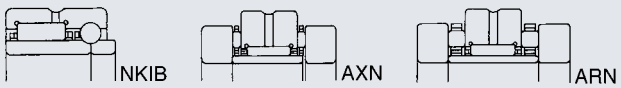
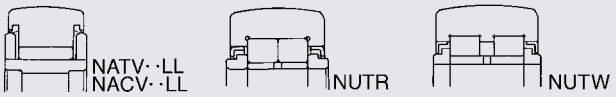
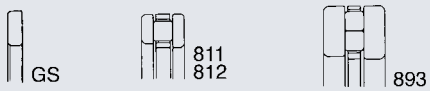
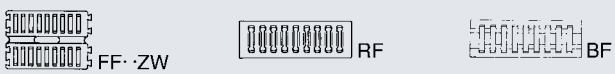
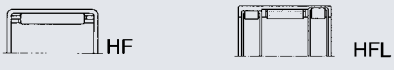

KLM


KH


KD


RLM


FF

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Warranty

NTN warrants, to the original purchaser only, that the delivered product which is the subject of this sale (a) will conform to drawings and specifications mutually established in writing as applicable to the contract, and (b) be free from defects in material or fabrication. The duration of this warranty is one year from date of delivery. If the buyer discovers within this period a failure of the product to conform to drawings or specifications, or a defect in material or fabrication, it must promptly notify **NTN** in writing. In no event shall such notification be received by **NTN** later than 13 months from the date of delivery. Within a reasonable time after such notification, **NTN** will, at its option, (a) correct any failure of the product to conform to drawings, specifications or any defect in material or workmanship, with either replacement or repair of the product, or (b) refund, in part or in whole, the purchase price. Such replacement and repair, excluding charges for labor, is at **NTN**'s expense. All warranty service will be performed at service centers designated by **NTN**. These remedies are the purchaser's exclusive remedies for breach of warranty.

NTN does not warrant (a) any product, components or parts not manufactured by **NTN**, (b) defects caused by failure to provide a suitable installation environment for the product, (c) damage caused by use of the product for purposes other than those for which it was designed, (d) damage caused by disasters such as fire, flood, wind, and lightning, (e) damage caused by unauthorized attachments or modification, (f) damage during shipment, or (g) any other abuse or misuse by the purchaser.

THE FOREGOING WARRANTIES ARE IN LIEU OF ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

In no case shall **NTN** be liable for any special, incidental, or consequential damages based upon breach of warranty, breach of contract, negligence, strict tort, or any other legal theory,and in no case shall total liability of **NTN** exceed the purchase price of the part upon which such liability is based. Such damages include, but are not limited to, loss of profits, loss of savings or revenue, loss of use of the product or any associated equipment, cost of capital, cost of any substitute equipment, facilities or services, downtime, the claims of third parties including customers, and injury to property. Some states do not allow limits on warranties, or on remedies for breach in certain transactions. In such states, the limits in this paragraph and in paragraph (2) shall apply to the extent allowable under case law and statutes in such states.

Any action for breach of warranty or any other legal theory must be commenced within 15 months following delivery of the goods.

Unless modified in a writing signed by both parties, this agreement is understood to be the complete and exclusive agreement between the parties, superceding all prior agreements, oral or written, and all other communications between the parties relating to the subject matter of this agreement. No employee of **NTN** or any other party is authorized to make any warranty in addition to those made in this agreement.

This agreement allocates the risks of product failure between **NTN** and the purchaser. This allocation is recognized by both parties and is reflected in the price of the goods. The purchaser acknowledges that it has read this agreement, understands it, and is bound by its terms.

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NTN

Needle Roller Bearings

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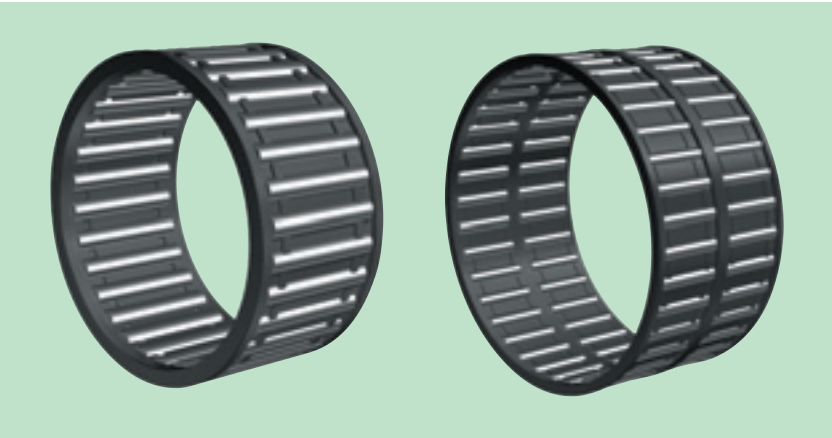
Appendix ····· C- 1

1. Classification and Characteristics Bearings

Needle roller bearings have relatively smaller diameter cylindrical rolling elements whose length is much larger than their diameter.

Compared with other types of rolling bearings, needle roller bearings have a small cross-sectional height and significant load-bearing capacity and rigidity relative to their volume. Also, because the inertial force action on them is limited, they are ideal choice for oscillating motion. Needle roller bearings contribute to compact and lightweight machine designs. They serve also as a ready replacement for sliding bearings.

NTN offers the following types of needle roller bearings.



This assembly, a major component of a needle roller bearing, comprises needle rollers and a cage to support the rollers.

- Using the shaft and housing as raceway surfaces reduces the cross-sectional height: it is equal to the diameter of the needle roller bearing.
- This structure eliminating the outer and inner rings allows the bearing to be fitted more easily.
- The assembly is available in both single-row and double-row configurations.
- As long as the tolerance limit of the shaft and housing is satisfied, the radial internal clearance can be made adjustable.

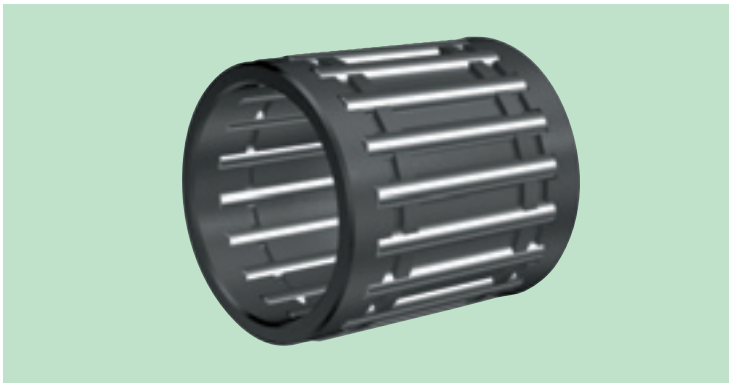
Needle roller and cage assembly for connecting rods

This assembly, a major component of a needle roller bearing, comprising needle rollers and a cage to support the rollers, is used typically for connecting rods in reciprocating compressors and small- and mid-sized internal combustion engines such as those for motorbikes, light cars, outboard motors and versatile engines. This assembly features such a cage that is specifically optimized for severe operating conditions involving high impact loads, complicated motions, high-speed revolution and/or high operating temperatures.

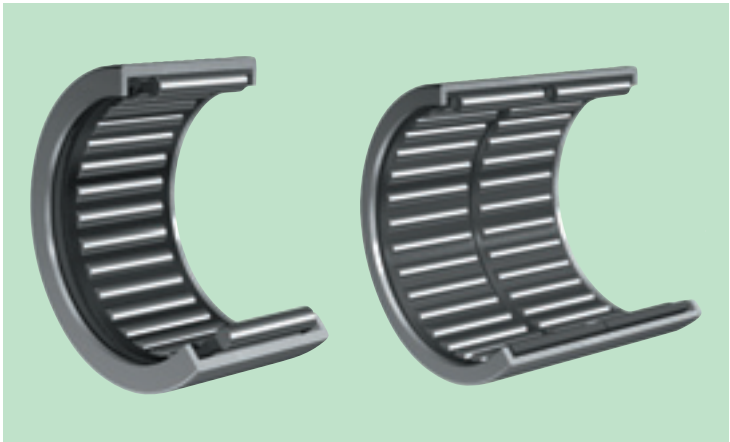
Needle roller and cage assembly for large end



- This assembly, subjected to a cranking motion with a simultaneous action of the rolling elements' rotation and revolution, must be light but have a high rigidity, and must have a precise dimension of the outer diameter of the cage so that the guiding system can keep an appropriate gap.
- The cage is made of high-tensile special steel with a surface hardening treatment.
- The guiding system employed is an outer diameter guiding system.
- The cage that may be subjected to poor lubrication can be protected with a surface treatment using a non-ferrous metal.
- For applications with a one-piece structure of crank shaft, the cage of split type is also available.



- Strong impact load acts on this bearing and it oscillates at high speed. Therefore, the cage bore surface is finished with high precision so as to secure lightweight, high rigidity and proper guide clearance.
- The cage is made of high tensile special steel and its surface is hardened ideally.
- The cage is of bore guide type and the guide surface is designed as long as possible to thereby reduce the surface pressure.
- The roller length is designed to the possible maximum value against the connecting width and a number of small-size needle rollers are fitted in the cage to minimize the contact pressure.



This bearing comprises an outer ring and needle rollers, which were both drawn from special thin steel plate by precision deep drawing, and a cage intended to guide precisely the needle rollers.

- This bearing is the type of the lowest section height, of the rolling bearings with outer ring, best-suited to space-saving design.
- A hardened and ground shaft or inner ring (IR Series) is used as the raceway.
- This bearing needs no axial locking due to easy installation and press-fit in the housing.
- The close end type to close shaft end is available in addition to the open end type.
- Furthermore, the type with seal fitted in at single side or double sides is also available.
- The standard type comprises a needle roller and cage assembly. In addition to this type, special type comprising full complement rollers is available at option.

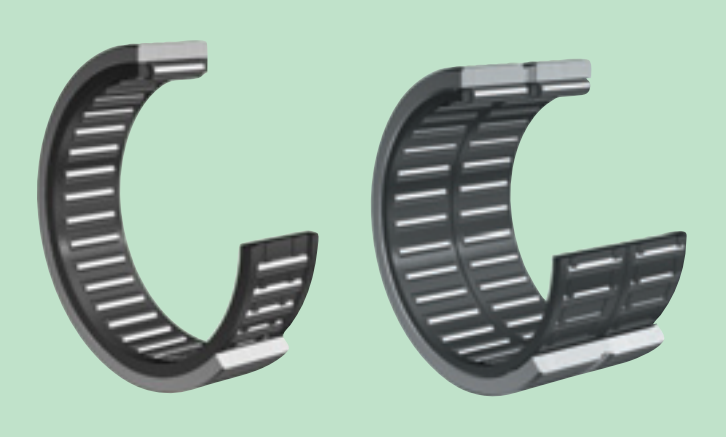
Machined-ring needle roller bearings



This bearing type comprises a machined outer ring and machined needle rollers, and a cage to guide properly the needle rollers.

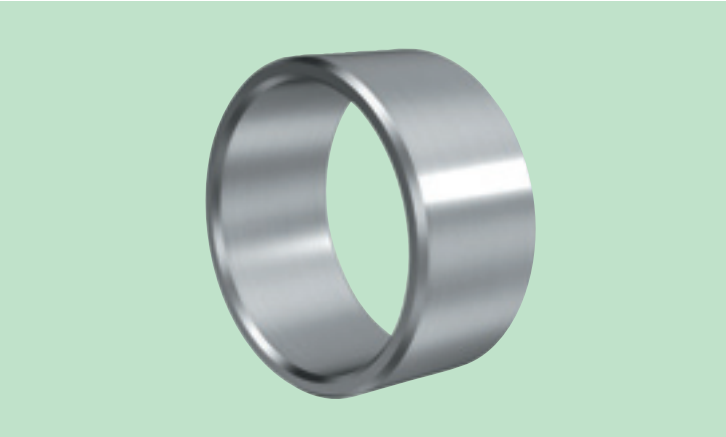
In the case of this bearing, the cage or the needle rollers are guided by the ribs of the outer ring or the face ring. Hence, this bearing is non-separable type. In addition, the type with no inner ring is also available for enabling a shaft to be used as the raceway surface. (Of course, the type with inner ring is available.)

- Selectively available for both of the metric system and the inch system.
- Best-suited to space-saving design due to its low section height, then having the large load capacity.
- High rigidity and high bearing accuracy due to its machined (precut) outer ring
- can be used with a housing made of light metal, etc. due to its outer ring of high rigidity.
- Outer ring with lubrication hole and lubrication groove
- Single row and double row types available.
- The type with seal fitted in at single side or double sides is also available.



This bearing type comprises a machined outer ring and machined needle rollers, and a cage to guide and retain properly the needle rollers. And the needle roller and cage assembly is separable from the outer ring. In addition to the standard type, the type with no inner ring is also available for enabling a shaft to be used as direct raceway surface. (Of course, the type with inner ring available, too.)

- Easy to install, — the needle roller and cage assembly, the outer ring, and the inner ring can be mounted independently from each other.
- Any optional radial internal clearance is selectable by combining the individual independent components.
- Best-suited to space-saving design due to its low section height, then having the large load capacity.
- High rigidity and high bearing accuracy due to its machined (precut) outer ring.
- Can be used with a housing made of light metal, etc. due to its outer ring of high rigidity.
- Single row and double row types available. And the outer ring of the double row bearing is provided with lubrication hole and lubrication groove.



For the needle roller bearings, usually a shaft is used as the raceway surface, but this inner ring is used where the shaft surface can not be machined to the specific hardness and roughness. This inner ring is suited to space-saving design due to its low section height. This is made of high carbon chrome bearing steel, finished by high precision grinding after heat-treated.

- Can also be used as a bush.
- Selectively available for both of the metric system and the inch system.
- The type with lubrication hole at the raceway center also available.

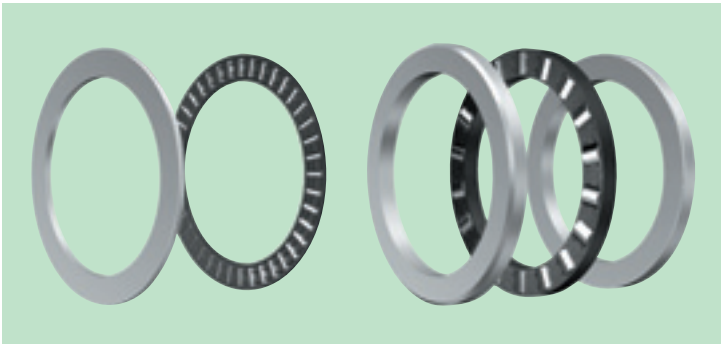
Clearance-adjustable needle roller bearing



This bearing type comprises a machined outer ring and machined needle rollers, and a cage to guide and retain properly the needle rollers, but the needle roller and cage assembly is non-separable from the outer ring. In addition to the standard type, the type with no inner ring is also available for enabling a shaft to be used as direct raceway surface. (Of course, the type with inner ring available, too.)

- The raceway diameter of the outer ring is shrunk by pressing the outer ring in axial direction, which then enables to reduce the roller set bore diameter correspondingly.
- Radial clearance is finely adjustable by adjusting axial pressing load and thereby changing shrinkage of the outer ring raceway diameter.
- This bearing is applied to the work spindle of a machine tool and other similar portions which require the high speed rotational accuracy of JIS Grade-4.

Thrust roller bearing



This bearing comprises needle rollers or cylindrical rollers, a cage to guide and retain properly the rollers, and disc-shaped rolling bearing ring. This is a bearing capable of supporting one-way axial load. Furthermore, this bearing can be used without rolling bearing ring, where the heat-treated and ground bearing mount surface can be used as the raceway surface.

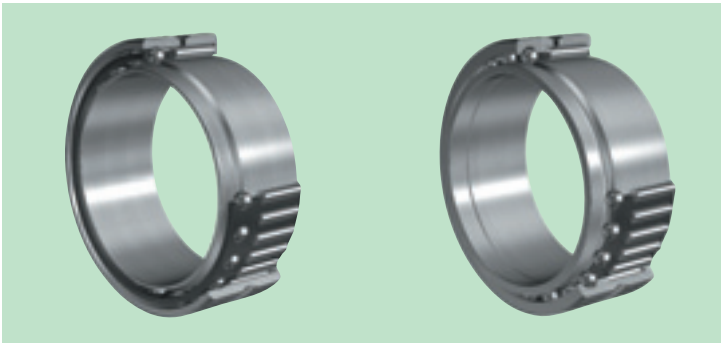
- Best-suited to space-saving design due to its small section height, then having the large load capacity.
- Available are AS type rolling bearing ring made of thin steel plate with hardened surface and WS type and GS type machined rings.



This complex bearing comprises a radial needle roller bearing for supporting radial load and a thrust bearing for supporting axial load which are assembled integrally. Both thrust ball bearing type and thrust axial roller bearing type are available as the bearing intended to support axial load.

- The thrust bearing with dust-proof cover is also available, which has a good effect in preventing scattering of oil and grease and invasion of external dust, etc.

Complex needle roller bearings
—Needle roller bearing with angular ball bearing, needle roller bearing with three-point contact ball bearing —



This complex bearing comprises a radial needle roller bearing for supporting radial load, a ball bearing for supporting comparatively small axial load and a machined inner ring which are all assembled integrally. Both angular ball bearing and three-point contact ball bearing are available as the ball bearing intended to support axial load.

- The complex needle roller bearings (NKIA Series)using an angular ball bearing as the thrust bearing can support one-directional angular load.
- The complex needle roller bearings (NKIB Series) using a three-point contact ball bearing as the thrust bearing can support double-directional axial load and furthermore its position in axial direction can be fixed, too.

Needle roller bearing with double thrust roller bearing

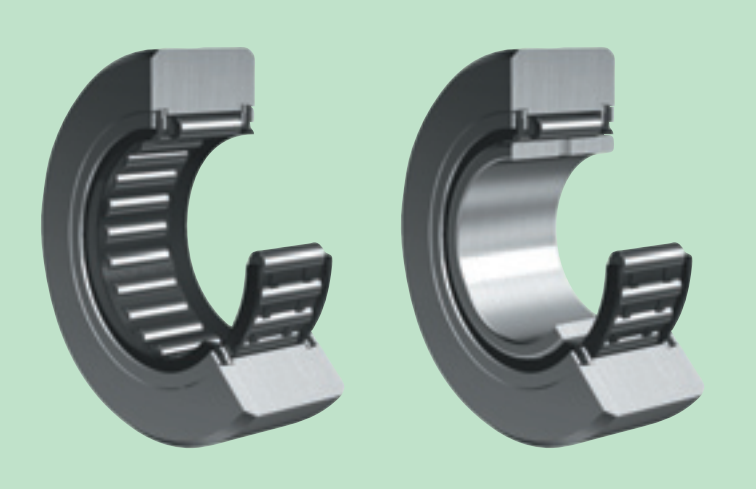


This is a complex bearing wherein a thrust needle roller bearing or a thrust cylindrical roller bearing intended to support axial load is configured at the double sides of a radial needle roller bearing for supporting radial load.

- Can support large axial load acting thereon from the double sides.
- Used as the bearing (precision bearing) for supporting the ball screw of a machine tool.

The track roller bearing is a needle roller bearing with thick outer ring, which is applied to cam roller, guide roller, eccentric roller and rocker arm.

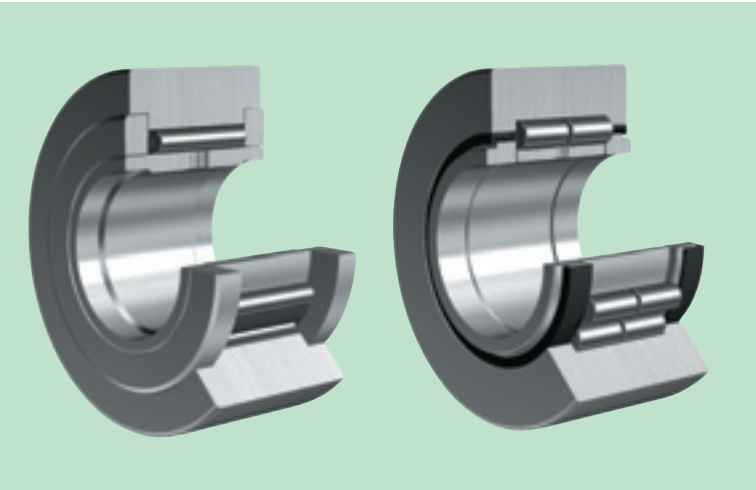
The track roller bearings are mainly categorized into a yoke type track roller bearing (roller follower) and a stud type track roller bearing (cam follower). Various types of the roller follower and the cam follower are available as described hereunder.



This roller follower is a bearing designed for rotation of the outer ring wherein a needle roller and cage assembly and a synthetic rubber seal reinforced with steel plate are assembled in the thick-walled outer ring.

- The outer ring, the needle roller and cage assembly, and the rubber seal are non-separable from each other.
- The outer ring is thick-walled type so that it is resistible to high load and impact load.
- The shaft must be provided with a thrust washer and a flange, because of the outer ring with no ribs (or face ring) and no axial guide function.
- The outer surface is available in both spherical (crowning) profile and cylindrical profile.
- The bearing with spherical outer ring is effective in damping bias load which is caused by deviation in installing.
- The bearing with cylindrical outer ring is suitable to the cases of large load and low-hardness track surface,due to its large area of contact with the mating track surface.

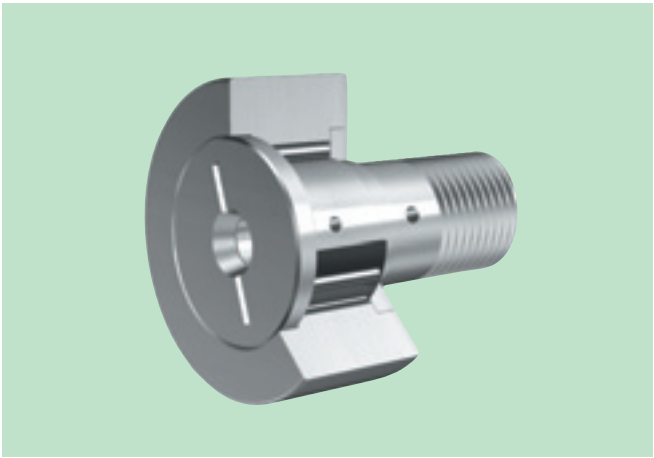
Roller follower —With axial guide—



This roller follower is a bearing designed for rotation of the outer ring wherein a needle roller and cage assembly, an inner ring, and a face ring are assembled in the thick-walled outer ring.

This bearing uses needle rollers as its rolling element and furthermore classified into bearing with cage and full complement roller bearing without cage. The outer ring is guided axially by a face ring which is press-fitted in the inner ring.

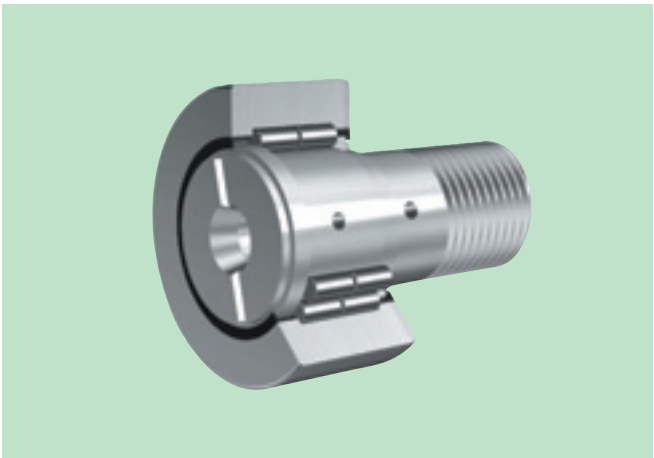
- The outer ring is thick-walled type so that it is resistible to high load and impact load.
- The outer surface is available in both spherical (crowning) profile and cylindrical profile.
- The bearing with spherical outer ring is effective in damping bias load which is caused by deviation in installing.
- The bearing with cylindrical outer ring is suitable to the cases of large load and low-hardness track surface,due to its large area of contact with the mating track surface.
- This bearing is easier to handle because it needs no mounting of a guide (thrust washer, etc.) on the shaft unlike other types without axial guide (RNA22, NA22).



This is a bearing designed for rotation of the outer ring in which a needle roller and cage assembly and a stud in lieu of inner ring are fitted in the thick-walled outer ring. The stud is so threaded as to be mounted easily. This cam follower (bearing) uses needle rollers as its rolling element and it is further classified into one bearing type with cage and another full complement roller bearing type without cage.

- The bearing type with cage is suitable to comparatively high speed running because its rollers are guided by the cage.
- The full complement roller type enabling to use more needle rollers than the type with cage has large load capacity.
- The outer surface is available in both spherical (crowning) profile and cylindrical profile.
- This cam follower (bearing) is selectively available for both of the metric system and the inch system.
- Furthermore, the seal built-in type is also available.
- The stud is of either recessed head type allowing use of a screwdriver or hexagon socket head type so as to be mounted and adjusted easily.

Cam follower —Cylindrical roller type—



This is a full complement roller bearing designed for rotation of the outer ring in which double-row cylindrical rollers and a stud in lieu of inner ring are fitted in the thick-walled outer ring. The stud is so threaded as to be mounted easily.

- This cam follower (bearing) has the radial /axial load capacity larger than the needle roller type.
- A steel plate is press-fitted in the outer ring and a labyrinth seal is formed between the face ring and the outer ring.
- The outer surface is available in both spherical (crowing) profile and cylindrical profile.
- The stud is of either recessed head type allowing use of a screwdriver or of hexagon socket head type so as to be mounted and adjusted easily.

Cam follower —Eccentric type—



This is a cam follower (bearing) wherein the studs of the needle roller type and cylindrical roller type prescribed above were made eccentric, which can then be adjusted by making eccentric the outer ring relative position against the raceway.

- Load distribution is easily adjustable in configuring two or more cam followers in linear form.
- Preload can be applied by adjustment of load distribution.
- Alignment is possible even when the mounting hole is not processed in high accuracy.
- The outer surface is selectively available in both spherical (crowning) profile and cylindrical profile.
- The stud is of either recessed head type allowing use of a screwdriver or hexagon socket head type so as to be mounted and adjusted easily.

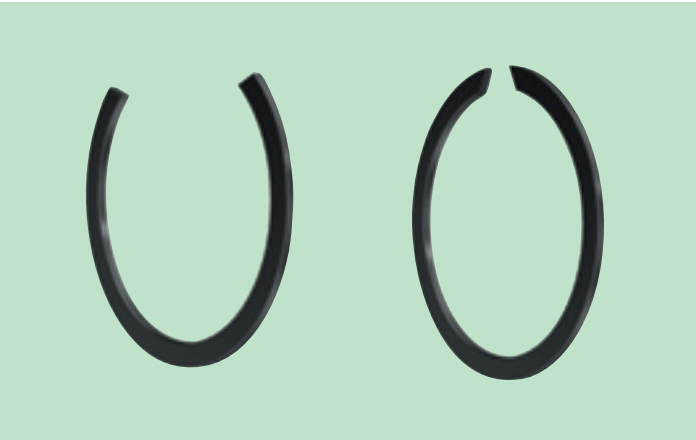
The components described hereunder are for needle roller bearing.



The needle rollers with flat end face and round end face are standard. These rollers are made of high-carbon chrome bearing steel, surface-finished by grinding and buffing after heat-treated. The standard accuracy of these rollers is "Precision Grade".

- A-Inter-diameter tolerance of the needle rollers is 2mm maximum.
- The roller type with crowned rolling surface is also available, which can damp edge load.
- These needle rollers are supplied as an individual for applications (pin, shaft) other than rolling element.

Snap rings



These are special-purposed rings used for axially positioning or guiding the inner and outer rings or the needle roller and cage assembly in needle roller bearing.

The ring material is a hard steel wire rod and chemical conversion treatment is applied to its surface so as to provide high rigidity.

- Two types are available for shaft and housing use.
- The section height is lowered flexibly according to the dimensions of needle roller bearings. In addition, manufacture of these rings is available up to the possible minimum dimensional range.
- For the axial guide it is recommended to provide a spacer between the cage and the snap ring.

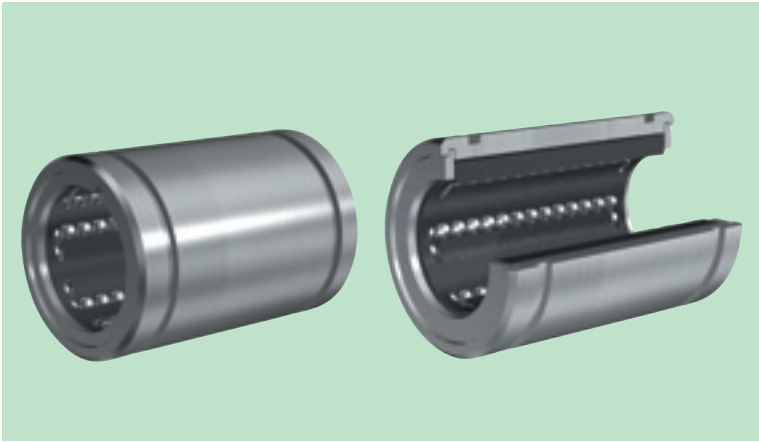
Seals



This is a special-purposed seal which was designed so as to match the small section height of needle roller bearings.

- G-type seal with one lip and GD-type seal with two lips are selectively available on application.
- These seals comprising steel ring and synthetic rubber can be used at the operating temperature ranging from -25 to 120˚C, but continuous use thereof is subject to 100˚C and less.
- These seals act to prevent invasion of external foreign matter and over-consumption of lubrication grease.
- The radial section height of each seal is designed so as to match the drawn-cup needle roller bearings. Hence, these seals require no additional finishing of the housing. This facilitates handling.

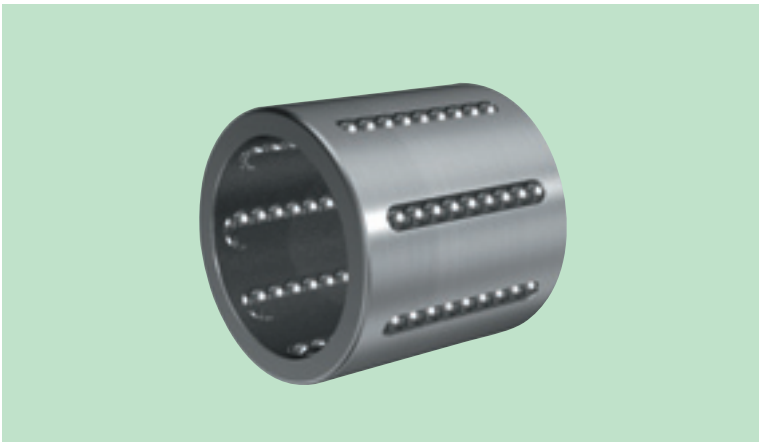
This catalogue describes the following ones of linear motion bearings.



This is a high precision bearing which comprises a machined outer ring and face ring, steel balls and a synthetic resin cage for retaining the balls and rolls on a shaft, maintaining the endless linear motion.

- Standard type, clearance-adjustable type and open type are selectively available on application.
- Some bearings of these types are provided with a synthetic rubber seal at single side or double sides to prevent invasion of foreign matter therein.
- The steel balls are guided precisely by the cage so that stable linear motion can be achieved with less friction resistance.
- No rotational motion is available.

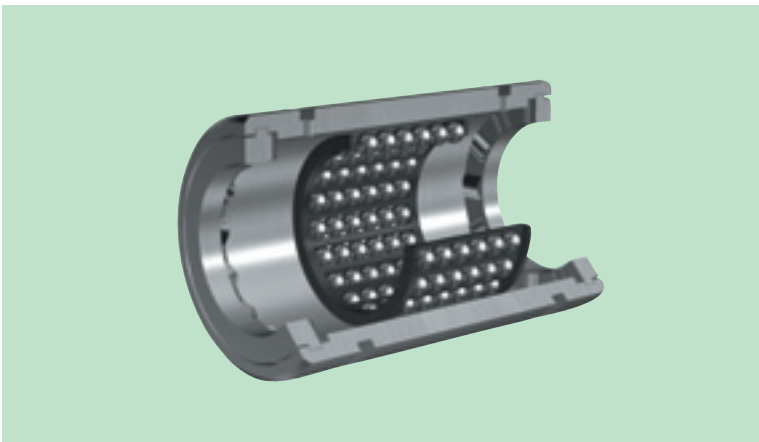
Linear ball bearing —Drawn cup type —



This is a high precision bearing which comprises an outer ring drawn from thin special steel plate by precision deep drawing, steel balls and a synthetic resin cage for retaining the balls and rolls on a shaft, maintaining the endless linear motion.

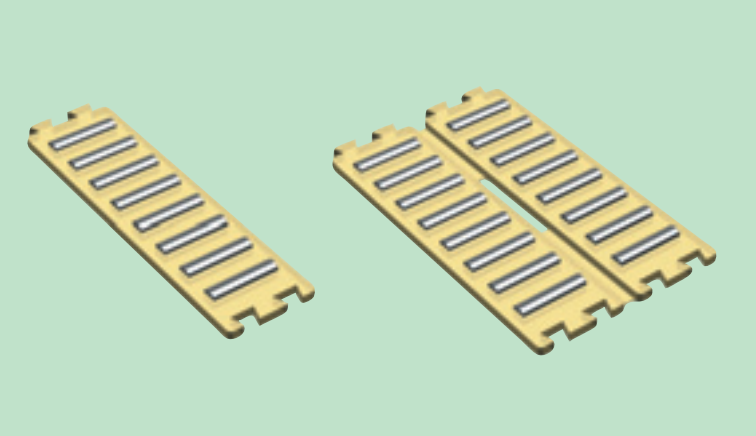
- The outer ring made of thin steel plate enables to make less the section height and to design the linear motion system of compact structure.
- Easy to install — This bearing is press-fitted in the housing so that it requires no axial fixing.
- No rotational motion available.
- Some bearings of this type are provided with a synthetic rubber seal at double sides to prevent invasion of foreign matter therein.

Linear ball bearing —Stroking type —



This bearing comprises a machined outer ring and face ring, steel balls and a cage for retaining the balls and reciprocates finitely on a shaft as well as rotates on a shaft. The outer ring is provided with a snap ring as the cage stopper at its double sides and, furthermore, a wavy spring is provided between the snap ring and the cage to damp impact acting the cage and to thereby prevent wear of the cage.

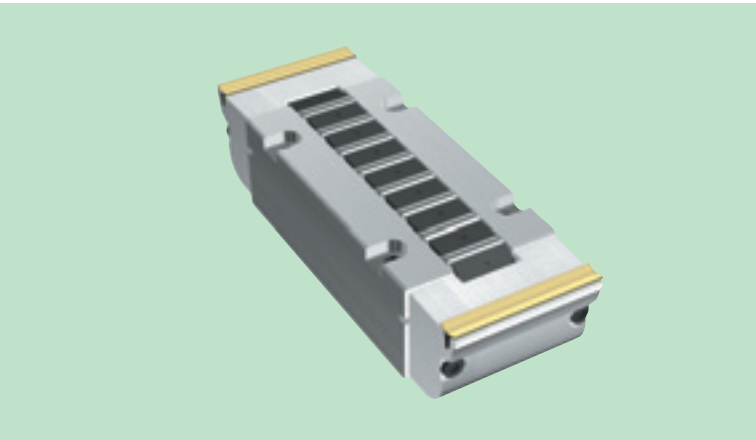
- Some bearings of this stroking type are provided with a synthetic rubber seal at its double sides to prevent invasion of foreign matter therein.
- The outer ring is so grooved that snap ring can be fitted and fixed easily.



This is a flat roller bearing which comprises a flat cage and needle rollers and reciprocates on a flat raceway in linear direction by motion of its linear movable components.

- The molded polyamide resin cage and press-formed steel plate cage are selectively available on application.
- FF type molded resin cage — Several cages can be used with them jointed with each other in parallel configuration.
- The press-formed steel plate cage — Cage to cage jointing is unavailable, but it can be supplied at any optional length.
- In the case of double-row resin cage, it is provided with an elastic joint at its center part so it can be reformed at any optional angle by dipping it in an oil of 70 to 90˚C and can be mounted on a Vee-grooved surface.

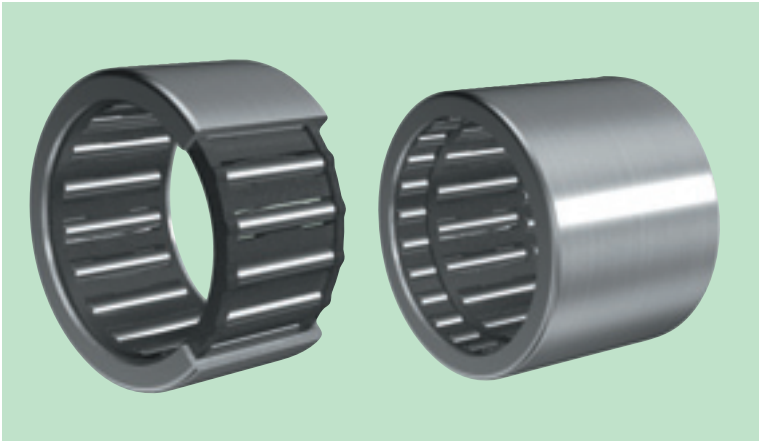
Linear roller bearing



This bearing comprising cylindrical rollers capable of turning internally around the track frame moves endlessly on a flat surface in linear direction.

- Low friction factor due to the case assemblies preventing neighboring roller from touching.
- High load rating due to use of cylindrical rollers.

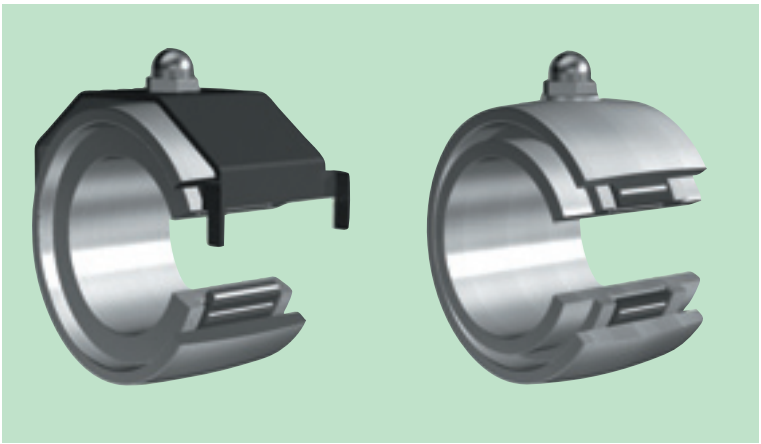
This catalogue describes the following products, too.



This one-way clutch comprises an outer ring drawn from thin special steel plate by precision deep drawing, spring, needle rollers and cage and can transmit a torque in only one way.

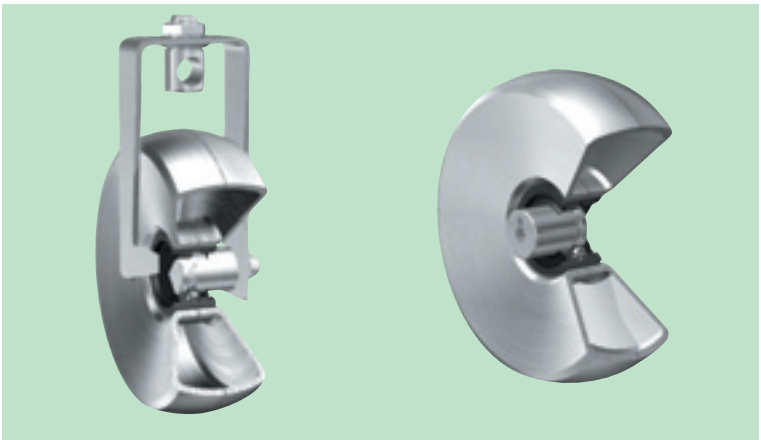
- Less friction torque against over-running, great transmission torque for small section height.
- Various types are available as follows; one-way clutch with bearing built therein to support radial load, one-way clutch with outer ring plated for improvement of erosion resistance, one-way clutch integrated with gear, pulley, etc.
- HF type and HFL type can be locked axially by being only press-fitted in the housing.
- These one-way clutches use the outer ring drawn by precision deep drawing, which needs the housing with wall thickness of a specified value or more.
- HF type is subjected to configuration of a radial ball bearing at its double ends. (On the other hand, HFL type is complete with radial bearings built-in at its double sides.)

Bottom roller bearing —For textile machinery—



This is a grease pre-filled needle roller bearing intended to support the bottom rollers. The spherical outer surface of the outer ring can allow bottom roller mounting error to some extent. On the other hand, the inner ring is provided with ribs at its double ends. A clearance between the outer ring ribs and the inner ring ribs is kept less and, in addition, the outer surface of each inner ring rib is knurled so as not to allow easy invasion of mist, etc. into the bearing.

Tension Pulley —For Textile Machinery—



This pulley is used to guide the tapered portion or belt of the spindle drive unit in each of a fine spinning machine, a rough spinning machine, a pre-twisting machine, etc. as well as to apply tension to them. The pulley drawn from steel plate by precision deep drawing is press-fitted on the outer ring of shaft bearing instead of inner ring.

2. Load Rating and Life

2.1 Bearing life

Even when bearings are in running under normal condition, the raceway surface and the surfaces of the rolling elements inevitably subjected to repetitive compression stress result in flaking due to fatigue of the raceway an element materials and can resist no longer to further running even under normal condition. "Bearing life" is defined as the total cumulative bearing revolutions until flaking occurs thus on the raceway surface or the element rolling surfaces.

Further use of bearings becomes unavailable, caused by various defects such as seizure, wear, crack, chipping, chattering, rusting, etc. in addition to the flaking phenomenon stated above. However, these phenomena should be deemed as defects of bearing itself and, therefore, be discriminated from bearing life. And the defect causes are such as incorrect selection of bearing, improper mounting, improper lubrication, imperfect sealing, etc. Hence, bearing trouble can be avoided by removing the said causes.

2.2 Basic rated life and basic dynamic load rating

Even when a group of same bearings have been in running under the same condition, considerably great variation is found among the individual bearing lives. This comes from variation in material fatigue itself. Hence, the basic rated life defined below considering statistically this variation in material fatigue is applied as the bearing life.

The basic rated life means the substantial total cumulative revolutions of the individual bearings 90% (90% reliability) of which can revolve without resulting in flaking due to rolling fatigue when a group of same bearings have been put individually in continuous running under same condition. When bearings have been in running in constant revolution, the basic rated life is expressed as the total cumulative revolutions of the individual bearings.

On the other hand, the basic dynamic load rating represents the load capacity of rolling bearing, then meaning a constant load under which the basic rated life of 1,000,000 revolutions is achieved. This constant load of radial bearing is expressed with net radial load, while that of thrust bearing is expressed with net axial load.

The bearing dimensions table in this catalogue describes the respective basic dynamic load ratings of the bearings that were manufactured using **NTN**'s standard materials and manufacturing processes in current use.

For the respective basic rated lives of the bearings manufactured using the special materials and special manufacturing processes, feel free to contact **NTN** for inquiry.

The relationship among the basic rated life, the basic dynamic load rating and the axial load can be expressed in **formula (2.1)**.

Basic Rated Life specified in ISO 281.

$$L_{10} = \left(\frac{C}{P}\right)^{p} \quad \cdots\cdots\cdots\

Table 2.1 Operating conditions and requisite lifetime

Operating conditions	Lifetime L_{10h}
Tools and devices requiring no all-time running Ex) Door switcher, etc.	500
Machines which are operated for a short time or intermittently and do not cause comparatively great affect on others even if they shut down incidentally due to trouble. Ex.) Hand tools, heavy material handling hoist in a machining shop, general hand-operated machines, agricultural machines, crane in a casting shop, material automatic feeder, home appliances, etc.	4 000 ～ 8 000
Machines which are not put in continuous running but required to run very precisely Ex.) Auxiliary machines in a power station, conveyor in a flow process, elevator, general cargo handling crane, machine tool of low frequency in use, etc.	8 000 ～ 14 000
Machines which are operated for 8 hours per day but not put in all-time continuous running Ex.) Power generator in a factory, general geared unit, etc.	14 000 ～ 20 000
Machines which are operated for 8 hours per day Ex.) General machines in a machining shop, crane in all-time operation.	20 000 ～ 30 000
Machines which are put in 24-hour continuous running Ex.) Separator, compressor, pump, main shaft, table roller of rolling mill, conveyor roller, winding engine in a mine, driving motor in a factory, etc.	50 000 ～ 60 000
Machines which are put in 24-hour continuous running and, in addition, are not allowed absolutely trouble shutdown. Ex.) Cellulose manufacturing machine, paper making machine, power station, drainage pump in a mine, urban city water related equipment, etc.	100 000 ～ 200 000

2.4 Bearing life using the adjusted life rating factor

The basic bearing life rating (90% reliability) can be calculated using the formulas prescribed in Subsection 2.2, but on occasion the bearing life of over-90% reliability must be determined on application. Also, the bearing life can be further extended by using a specially improved bearing material and special manufacturing process/technique. Furthermore, it was clarified by the elastohydrodynamic lubrication theory that the bearing life would be influenced by bearing operating conditions (lubrication, temperature, speed, etc.)

The bearing life considering the above factors can be determined by **formula (2.7)** using "Life Adjustment Factor" specified in ISO 281.

$$L_{na} = a_1 \cdot a_2 \cdot a_3 \ (C/P)^p \quad \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \textbf{(2.7)}$$

where,

L_{na} : Adjusted life rating 10^6 revolutions
a_1 : Reliability adjustment factor
a_2 : Bearing material adjustment factor
a_3 : Operating condition adjustment factor

2.4.1 Reliability adjustment factor a_1

The values of reliability adjustment factor a_1 for 90% and higher reliability are as shown in **Table 2.2**.

Table 2.2 Values of reliability adjustment factor a_1

Reliability %	L_n	Reliability adjustment factor a_1
90	L_{10}	1.00
95	L_5	0.62
96	L_4	0.53
97	L_3	0.44
98	L_2	0.33
99	L_1	0.21

2.4.2 Life adjustment factor for bearing material a_2

Where bearing material used is special type and quality and manufactured in the special process, the life-related bearing characteristics are inevitably variable depending on the specialties of the material. In such a case, the bearing life is adjusted using the life adjustment factor for bearing material a_2.

The basic dynamic load ratings described in "Bearing Dimensions Table" are subject to use of the standard materials and manufacturing processes / techniques being used in **NTN**, and usually a_2=1 rating is used.

Furthermore, a_2>1 is eventually applied to the bearings manufactured using the specially improved material and manufacturing process / technique. In such a case, feel free to contact **NTN** for further instruction.

Where bearings made of high carbon chrome bearing steel are used at 120˚C and over throughout a long term, the dimension of the normal heat-treated bearing varies significantly and, therefore, dimension-stabilizing heat-treatment (TS treatment) must be done to the bearings according to the maximum operating temperature. However, this dimension-stabilizing treatment causes the bearing hardness to reduce, finally an inverse affect on the bearing life. To avoid it, the bearing life is adjusted (compensated) by multiplying the life rating by the life adjustment factor for bearing material a_2 shown in **Table 2.3**.

Table 2.3 Values of life adjustment factor for bearing material a_2 for dimension-stabilizing heat-treated (TS-treated) bearings

Code	Maximum operating temperature	Life adjustment factor for bearing material a_2
TS2-	160˚C	1.00
TS3-	200˚C	0.73
TS4-	250˚C	0.48

2.4.3 Life adjustment factor for operating conditions a_3

The life adjustment factor for operating conditions a_3 is used for adjustment(compensation) of the bearing life, where the lubricated bearing condition gets worse and the lubricant deteriorates or foreign matter is included in the bearing, caused by the rotational speed, temperature rise, etc. of the bearing in running.

Generally the life adjustment factor in the case of good lubricated bearing condition is a_3=1 and, particularly where the lubricated bearing condition is good and other factors for the bearing are normal, a_3>1 can be applied to adjustment of the bearing life. However, a_3<1 is applied in the following cases.

- **Low viscosity of lubrication oil under bearing** temperature in running
  Radial needle roller bearing 13mm²/s and less
  Thrust needle roller bearing 20mm²/s and less
- **Particularly low rotational speed**
  (The product of rotational speed n r/min by pitch circle diameter d_p mm of rolling element is $d_p \cdot n$<10000.)
- **High operating temperature of bearing**
  High operating temperature of bearing would cause the raceway hardness to reduce, whereby bearing life would be shortened, In such a case, the bearing life is adjusted (compensated) by multiplying the value shown in **Fig.2.1** as the life adjustment factor for operating conditions depending on operating temperature.
- **Inclusion of foreign matter in lubricant**
  In the case of special operating conditions, feel free to contact **NTN** for inquiry. For a bearing manufactured using the specially improved material and manufacturing process/technique, usually $a_2 \times a_3$<1 is applied unless the lubricating condition is good, even though the life adjustment factor is a_2>1.

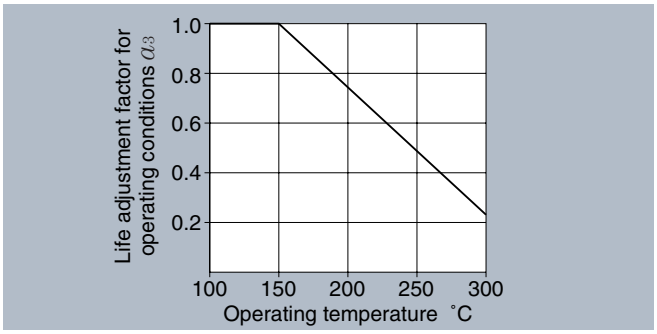


Fig. 2.1 Life adjustment factor for operating conditions depending on operating temperature

2.5 Affect on basic dynamic load rating by raceway surface hardness

When a bearing rolls on shaft surface/housing surface as its raceway surface, the surface layer must be hardened to HRC58 to64 in proper hardening depth.

Ordinary quenching, carburizing or induction quenching, etc. are available as the raceway hardening method, but low hardness of the raceway would cause to the bearing fatigue lifetime to reduce. In such a case, the basic load rating is adjusted by multiplying the hardness factor shown in **Fig.2.2**.

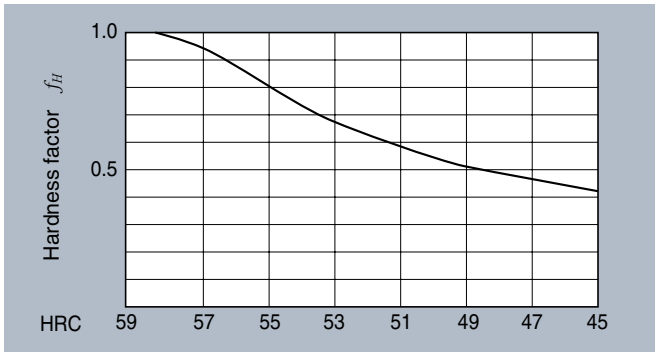


Fig. 2.2 Hardness factor

2.6 Life of bearing with oscillating motion

The life of bearing with oscillating motion can be determined by **formula (2.8)**.

$$L_{osc} = \Omega L_{Rot} \cdots\cdots (2.8)$$

where,

L_{osc} : Life of bearing with oscillating motion
L_{Rot} : Life rating subject to rotational speed r/min identical to oscillation frequency cpm
Ω : Oscillation factor (showing the relation with half angle β of oscillation angle per **Fig.2.3**).

Generally **Fig.2.3** is applied when the oscillation angle exceeds critical angle 2β. This critical angle is nearly determined by design of bearing internals —mainly the number of rolling elements included in single row bearing.

Where a bearing with oscillating motion is used at below the critical angle, its life is shorter than the lifetime value calculated using **Fig.2.3**. When the critical angle is unknown, determine Ω, assuming $\beta = \beta_C$. Feel free to contact **NTN** for inquiry about the critical angle of individual bearings.

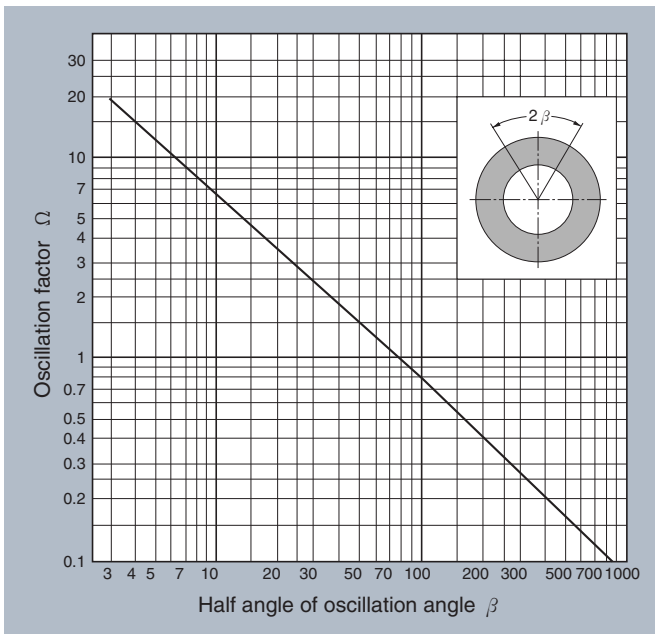


Fig. 2.3 Relationship of oscillation angle β to factor Ω

When the oscillation angle 2β is very small, difficulty in forming an oil film on the contact surface of rolling ring to rolling element could result in fretting corrosion.

In the case of inner ring oscillation, the critical oscillation angle is expressed in **formula (2.9)**.

$$\text{Critical oscillation angle } \theta \geqq \frac{360^\circ}{Z} \cdot \frac{d_p}{d_p - D_a \cos\alpha} \quad \cdots\cdots \textbf{(2.9)}$$

Where,

Z : Number of rolling elements (per row)

d_p: Pitch circle diameter (PCD) of rolling element

D_p: Rolling element diameter

e : Contact angle

(In the case of outer ring oscillation, the right side denominator is $d_p + D_a \cos\alpha$.)

2.7 Life of bearing with linear motion

In the case of bearings with linear motion such as linear ball bearing, linear flat roller bearing, etc., the relationship among axial travel distance, bearing load and load rating can be expressed in **formulas (2.10)**, **(2.11)**.

When the rolling elements are balls;

$$L = 50 \times \left(\frac{C_r}{P_r}\right)^3 \quad \cdots\cdots \textbf{(2.10)}$$

When the rolling elements are rollers;

$$L = 100 \times \left(\frac{C_r}{P_r}\right)^{10/3} \quad \cdots\cdots \textbf{(2.11)}$$

where,

L : Load rating km

C_r : Basic dynamic load rating N (kgf)

P_r : Bearing load N (kgf)

Fig.2.4 shows the relationship of C_r/P_r to L.

Furthermore, when the travel motion cycle and travel distance remain unchanged, the lifetime of bearing can be determined by **formulas (2.12)**, **(2.13)**.

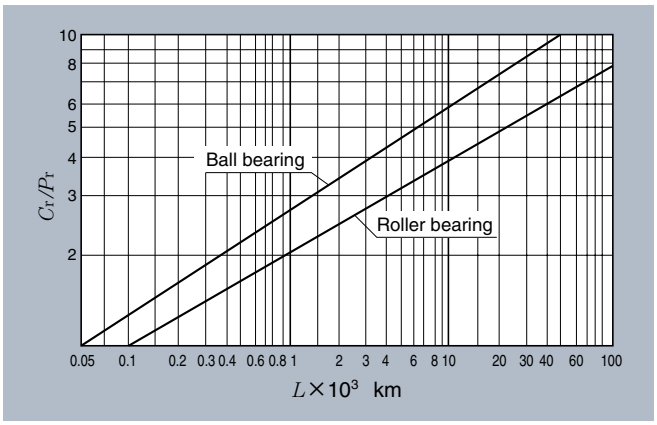


Fig. 2.4 Life of bearing with axial motion

When the rolling elements are balls;

$$L_h = \frac{50 \times 10^3}{60 \cdot S} \left(\frac{C_r}{P_r}\right)^3 \quad \cdots\cdots \textbf{(2.12)}$$

When the rolling elements are rollers;

$$L_h = \frac{100 \times 10^3}{60 \cdot S} \left(\frac{C_r}{P_r}\right)^{10/3} \quad \cdots\cdots \textbf{(2.13)}$$

where,

L_h: Travel life h

S : Travel distance per minute m/min

$S = 2 \cdot L \cdot n$

L : Stroke length m

n : Stroke cycle N (kgf)

2.8 Various factors influencing bearing life

Bearing life is influenced by not only bearing load and rotational speed but also various factors such as lubricating condition, internal clearance, surface roughness and hardness of raceway, heat-treated structure, misalignment, etc. In using bearings, therefore, these influence factors must be taken into full consideration. **Table 2.4** shows the bearing operating conditions as a guideline. (For the detail refer to "Commentary" described in Bearing Dimensions Table every each bearing type.)

Table 2.4 Guideline of the bearing operating conditions

Allowable rotational speed (r/min)	See Dimensions Table
Surface roughness of raceway	within 0.2a
Surface hardness of raceway	HRC58 to 64 Note: Refer to Subsection 7.5 for various materials and heat-treated hardness thereof.
Misalignment	1/2000 or less (for radial bearing)
Radial internal clearance	Ordinary level (C2, C3, C4)

2.9 Fitting misalignment and crowning

Generally it is well known that stress concentration at edge portion (so called, edge load) arising from fitting misalignment would result in rapid reduction of bearing lifetime. "Crowning" is adopted as a countermeasure against such rapid reduction of bearing lifetime. In that case, however, unless it is designed properly this crowning would cause the effective contact length of roller to reduce, which could then lead to shorter life of bearing. It is therefore necessary to calculate a proper crowning value though depending on the extent of fitting misalignment.

And load condition. For reference, **Figs.2.5** to **2.7** show the examples of contact surface pressure analysis by a computer.

As seen from **Figs.2.5** to **2.7** (Examples of contact surface pressure analysis), the rollers with no crowning have high edge surface pressure, while the rollers with crowning restrain the edge surface pressure lower in the range of specific allowable fitting misalignment. **Fig.2.8** shows the relationship of allowable fitting misalignment to bearing lifetime. (example of computer analysis) It is possible to see from this Figure how the bearing lifetime is influenced by fitting misalignment.

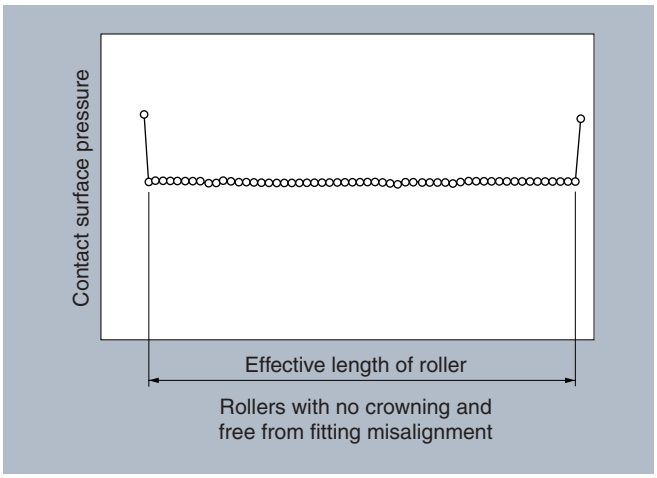


Fig. 2.5

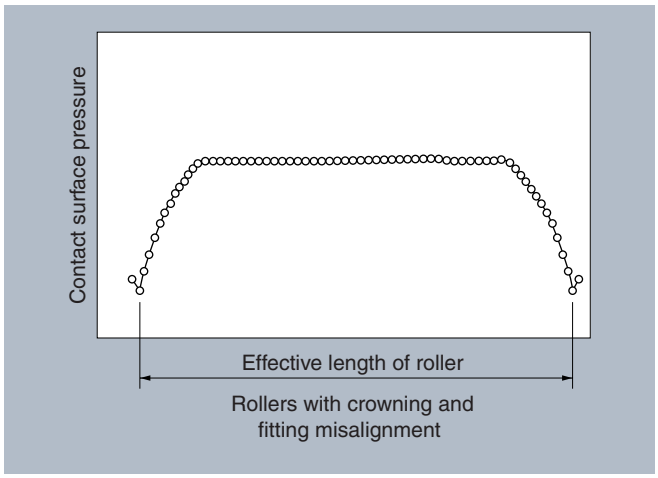


Fig. 2.7

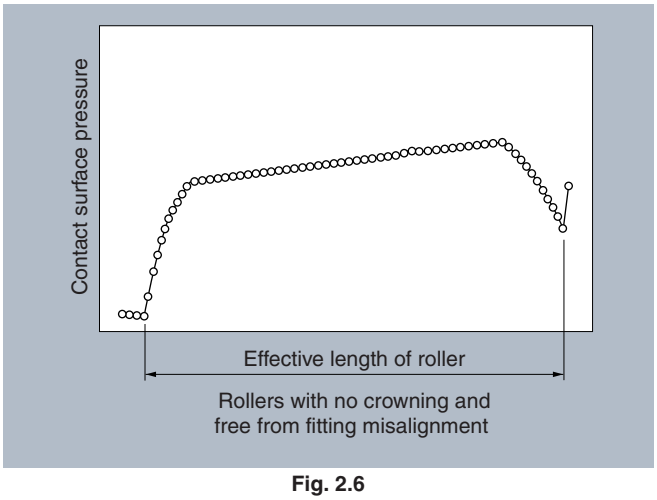


Fig. 2.6

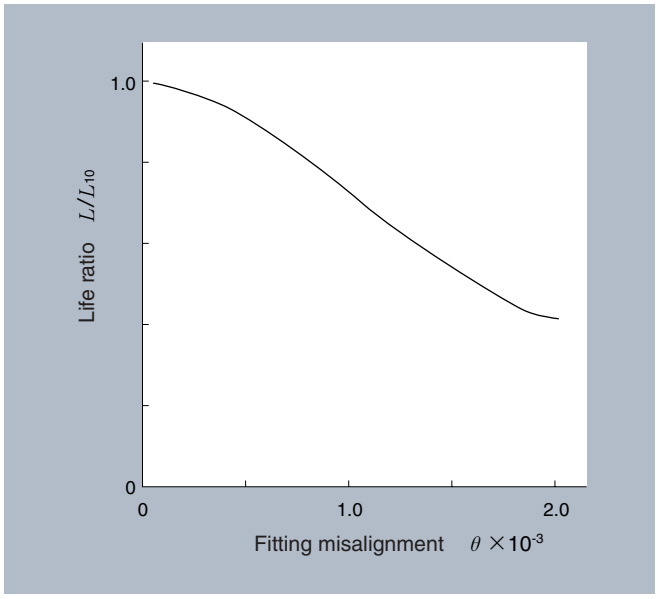


Fig. 2.8 Relationship of fitting misalignment to bearing lifetime

2.10 Radial internal clearance, surface roughness and surface hardness, and bearing life

The relationship of radial internal clearance to bearing life is as shown in **Fig. 2.9** and the relationship of surface roughness to bearing life as shown in **Fig.2.10**.

It is possible to see from these Figures how the bearing life is influenced by each factor.

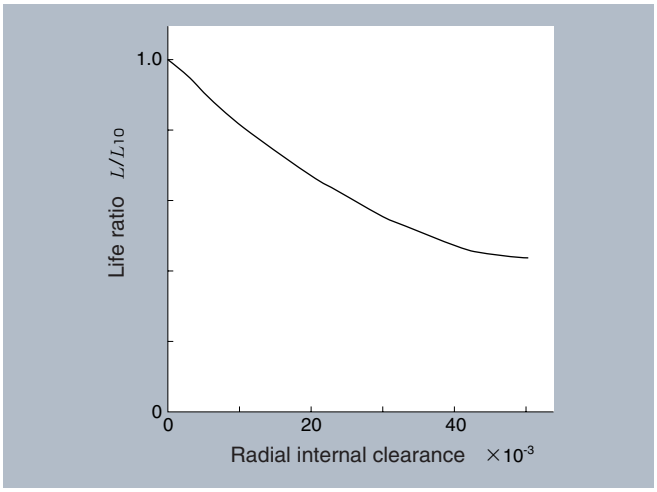


Fig. 2.9 Relationship of radial internal clearance to bearing life

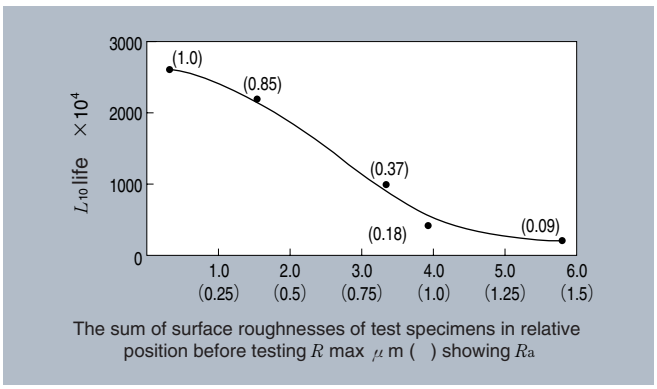


Fig. 2.10 Relationship of surface roughness to bearing life

2.11 Basic static load rating

Load acting on a bearing, when acted, results in locally permanent deformation of the contract surface of rolling elements to bearing ring. And this deformation value increases inevitably with the increasing load and smooth rotation of the bearing is interfered with by the deformation when it exceeded a certain limit value.

It is known experimentally that the total permanent deformation value 0.0001 times as large as the rolling element diameter at the contact center of rolling elements to raceway both of which are subjected to maximum stress is an allowable deformation limit which does not interfere with smooth rotation of bearing.

"Basic static load rating" is defined as such a constant static load that results in permanent deformation of the said limit value, which is then expressed in net radial load for radial bearings and in net axial load for thrust bearings.

Regarding this load value, each table of bearing dimensions describes it in the C_{or}, field for radial bearings and in C_{oa} field for thrust bearings respectively.

When the load defined above acts on a bearing, contact stress at the contact center of rolling element to raceway, which are subjected to maximum load, reaches the following value.

For roller bearing ·····4000MPa (408kgf/mm²)
For ball bearing ·····4200MPa (428kgf/mm²)

2.12 Allowable static bearing load

The basic static load rating prescribed in Subsection 2.11 is generally deemed as an allowable static bearing limit load, but in some cases this allowable limit load is set up larger than the basic static load rating and in some other cases it is set up smaller, according to the requirements for revolving smoothness and friction.

Generally this allowable limit load is decided considering the safety factor So in the following **formula (2.14)** and **Table 2.5**.

$$S_o = C_o / P_o \cdots\cdots (2.14)$$

where,
S_o : Safety factor
C_o : Basic static rated load, N (kgf)
(For radial bearings: C_{or},
For thrust bearings: C_{oa})
$P_{o\,max}$: Maximum static bearing load, N (kgf)
(For radial bearings: P_{or} max,
For thrust bearings: C_{oa} max)

Table 2.5 Lower limit value of safety factor S_o

Operating conditions	Roller bearings	Ball bearings
Requirement for high revolving accuracy	3	2
Requirement for ordinal revolving accuracy (ordinary-purposed)	1.5	1
Where minor deterioration of revolving accuracy is allowed (Ex. Low speed revolution, duty load application, etc.)	1	0.5

Remarks: 1. For the drawn-cup needle roller bearings, 3 shall be adopted as S_o lower limit value.
2. Where vibration and shock load act on bearing, P_o max shall be determined considering the shock load factor.

3. Calculation of Bearing Loads

3.1 Load acting on shafts

The loads acting on the shaft to be supported with the bearing must be determined for calculation of bearing loads. These loads include the self-weight of rotor assy, loads inevitably resulting from a machine in running, loads arising from power transmission, etc. These loads are mostly difficult to calculate though some of them can be calculated numerically and theoretically.

The following paragraphs describe how to calculate the loads acting on a power transmission shaft as one of the main bearing applications.

3.1.1 Load acting on gears

The loads acting on gears can be divided into tangential load (K_t), radial load (K_s) and axial load (K_a). The magnitude and acting direction of each load differ depending on the types of gear. This paragraph describes how to calculate the loads acting on parallel shaft gears and cross shaft gears for general use. For calculation of the loads acting on other gears, feel free to contact **NTN** for inquiry.

(1) Load acting on parallel shaft gear

Figs. 3.1 to **3.3** illustrate the loads acting on spur gear and helical gear which are used with a parallel shaft. The magnitude of each load can be determined using the **formulas (3.1)** to **(3.4)**.

$$K_t = \frac{19.1\times10^6 \cdot HP}{D_p \cdot n}\left(\frac{1.95\times10^6 \cdot HP}{D_p \cdot n}\right) \quad \cdots\cdots (3.1)$$

$$K_s = K_t \cdot \tan\alpha \text{ (Spur gear)} \quad \cdots\cdots (3.2a)$$

$$= K_t \cdot \frac{\tan\alpha}{\cos\beta} \text{ (for helical gear)} \quad \cdots\cdots (3.2b)$$

$$K_r = \sqrt{K_t^2 + K_s^2} \quad \cdots\cdots (3.3a)$$

$$K_a = K_t \cdot \tan\beta \text{ (for helical gear)} \quad \cdots\cdots (3.4)$$

where,

K_t : Tangential load acting on gear (Tangential force) N (kgf)
K_s : Radial load acting on gear (separating force) N (kgf)
K_r : Load acting perpendicularly on gear shaft (composite force of tangential force and separating force) N (kgf)
K_a : Parallel load acting on gear shaft N (kgf)
HP : Transmission power kw
n : Rotational speed r/min
D_p : Pitch circle diameter of gear mm
α : Gear pressure angle
β : Gear helix angle

Actual gear load is determined by multiplying the theoretical load by the gear factor described in **Table 3.1**, because vibration and shock are added to each theoretical load determined by the above formulas.

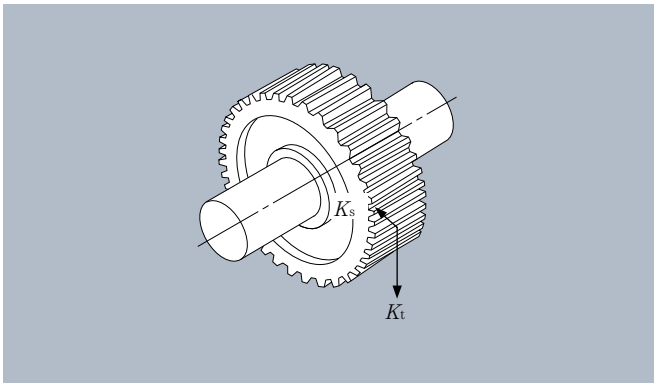


Fig. 3.1 Load acting on spur gear

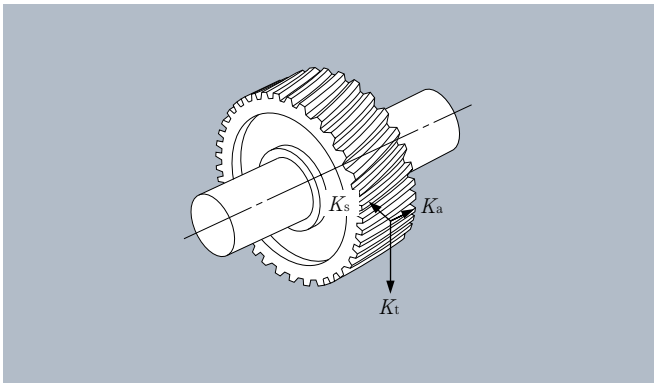


Fig. 3.2 Load acting on helical gear

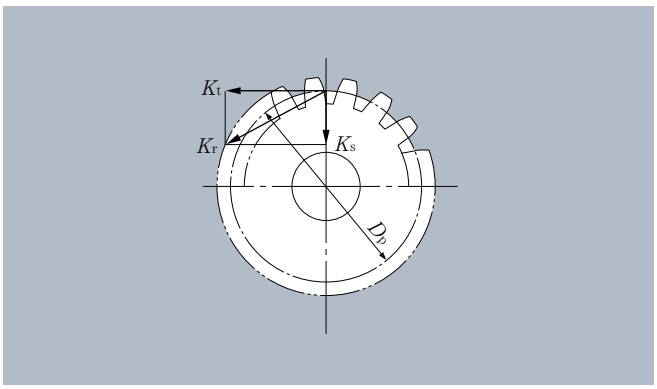


Fig. 3.3 Composite radial force acting on gear

Table 3.1 Gear factor f_z

Types of gear	f_z
Precision ground gears (Pitch and profile errors of less than 0.02mm)	1.05～1.1
Ordinary machined gears (Pitch and profile errors of less than 0.1mm)	1.1～1.3

(2) Loads acting on cross shaft gears

Figs. 3.4 and *3.5* illustrate the loads acting on straight-too bevel gears and spiral bevel gears which are used with cross shafts.

Table 3.2 shows how to calculate these loads.

The load acting on straight-tooth bevel gear can be determined from **Table 3.2**, assuming the helix angle as β =0.

The symbols and units used in this table are as follows.

K_t : Tangential load acting on gear (Tangential force) N (kgf)
K_s : Radial load acting on gear (separating force) N (kgf)
K_a : Parallel load acting on gear shaft (axial load) N (kgf)
HP : Transmission power kw
n : Rotational speed r/min
D_{pm} : Mean pitch circle diameter mm
α : Gear pressure angle
β : Gear helix angle
δ : Pitch cone angle of gear

In general, the relationship between the loads acting on pinion and pinion gear can be expressed as follows, due to the perpendicular intersection of two shafts.

$$K_{sp}=K_{ag} \quad \cdots\cdots (3.5)$$
$$K_{ap}=K_{sg} \quad \cdots\cdots (3.6)$$

where,

K_{sp}, K_{sg} : Pinion and pinion gear separating force N (kgf)
K_{ap}, K_{ag} : Axial load acting on pinion and pinion gear N (kgf)

For spiral bevel gears, the loading direction differs depending on the direction of helix angle, rotational direction and whether the spiral bevel gear is located at the drive side or the driven side.

The separating force (K_s) acting direction and axial load (K_a) acting direction illustrated in **Fig.3.5** are the positive direction respectively. On the other hand, the rotational direction and helix angle direction are to be defined viewing from the large end side of the gear. Hence, for the gear illustrated in **Fig. 3.5** these directions are clockwise and to right.

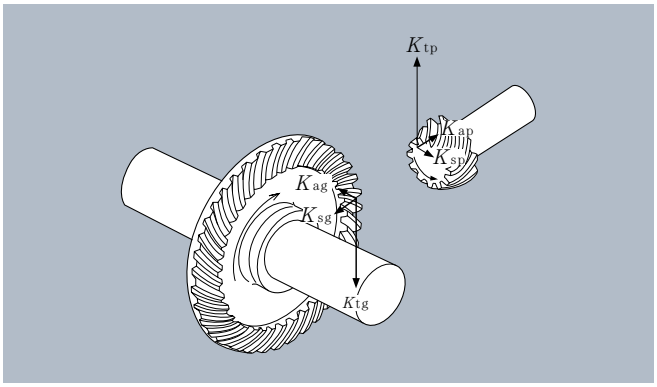


Fig. 3.4 Load acting on bevel gears

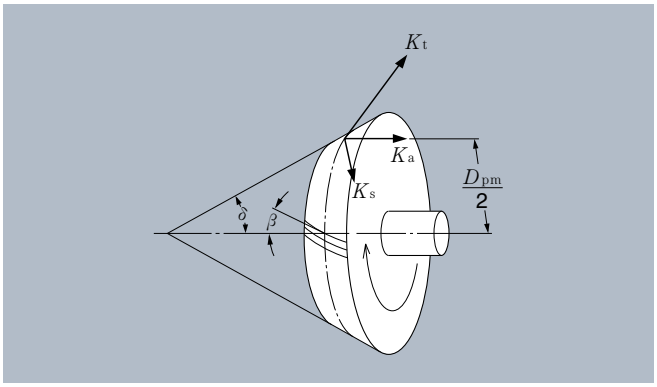


Fig.3.5 Bevel gear diagram

Table 3.2 Loads acting on bevel gears

Unit N

Driving gears	Rotational direction	Clockwise	Counter clockwise	Clockwise	Counter clockwise
	Helix angle	To right	To left	To left	To right
Tangential load K_t		$K_t=\frac{19.1\times10^6\cdot H}{D_{pm}\cdot n}$, $\left\{\frac{1.95\times10^6\cdot H}{D_{pm}\cdot n}\right\}$			
Separating force K_s	Drive side	$K_s=K_t\left(\tan\alpha\frac{\cos\delta}{\cos\beta}+\tan\beta\sin\delta\right)$		$K_s=K_t\left(\tan\alpha\frac{\cos\delta}{\cos\beta}-\tan\beta\sin\delta\right)$	
	Driven side	$K_s=K_t\left(\tan\alpha\frac{\cos\delta}{\cos\beta}-\tan\beta\sin\delta\right)$		$K_s=K_t\left(\tan\alpha\frac{\cos\delta}{\cos\beta}+\tan\beta\sin\delta\right)$	
Axial load K_a	Drive side	$K_a=K_t\left(\tan\alpha\frac{\sin\delta}{\cos\beta}-\tan\beta\cos\delta\right)$		$K_a=K_t\left(\tan\alpha\frac{\sin\delta}{\cos\beta}+\tan\beta\cos\delta\right)$	
	Driven side	$K_a=K_t\left(\tan\alpha\frac{\sin\delta}{\cos\beta}+\tan\beta\cos\delta\right)$		$K_a=K_t\left(\tan\alpha\frac{\sin\delta}{\cos\beta}-\tan\beta\cos\delta\right)$	